جامعة المرقب
كلية تقنية المعلومات

Object Oriented Programming

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02

المحاضرة الثانية

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The Outline

01 Reviews of class and objects
02 Examples
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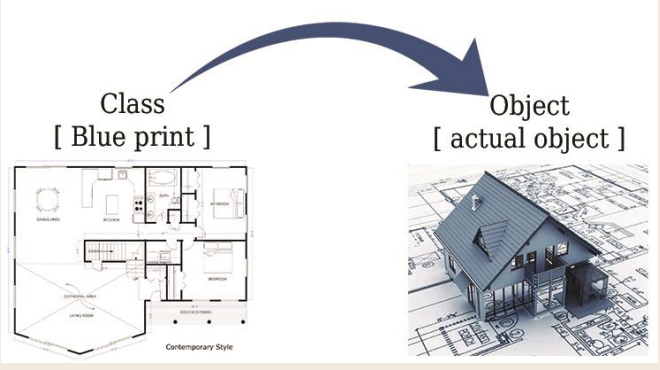
Reviews of class and objects

Class: Blueprint or template for creating objects
Object: Instance of class
Attributes: Data/variables that store information
Methods: Functions/behaviors that perform actions



Reviews of class and objects

Class: Blueprint or template for creating objects
Object: Instance of a class
Attributes: Data/variables that store information
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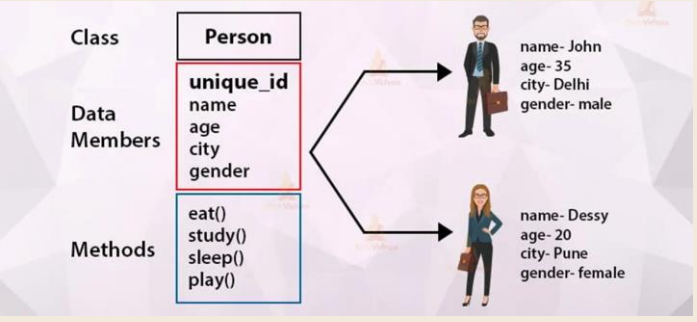
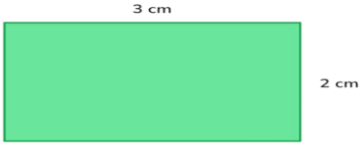


Example1 calculate the area of a rectangle using a class. The program should:
set values to the attributes(data) in main function .**area=Length*Width**

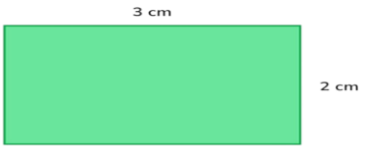
Step 1: Identify Attributes and Methods
we need to think: What describes a rectangle(information)? What can it do?"

Attributes (Data):	Methods (Functions):
Length=3.0	CalculateArea
Width =2.0	



Step 2: follow General Steps for Creating a Class and its objects
1-Declare Class
2. Define Attributes
3. Define Methods
4.Create Object
5. Set Attribute Values(or read by user)
6. Call Methods

Example1 calculate the **area** of a rectangle using a class. The program should:
set value to the attributes(data) in main function




```
#include <iostream>
using namespace std;

//1-Declare Class
class Rectangle
 {
//2-Define Attributes

public:
   double length;
   double width;
// 3-Define Methods

 void calculateArea() {
        cout << "Area: " << (length * width) << endl;
   }
};

int main() {

// 4-Create object
Rectangle rect1;

//5-Set Attribute Values(assign or read by user)

   rect1.length = 3.0;
   rect1.width = 2.0;

 //6. Call Methods

rect1. calculateArea() ;

}
```

Example 2:calculate the area of a rectangle using a class. The program should:
use public methods to read input values and calculate area in class

Attributes (Data):	Methods (Functions):
Length	ReadData()
Width	CalculateArea

```
#include <iostream>
using namespace std;
//1-Declare Class
class Rectangle
 {

//2-Define Attributes

public:
   double length;
   double width;
// 3-Define Methods

void readData() {
      cout << "Enter length of rectangle: ";
      cin >> length;
      cout << "Enter width of rectangle: ";
      cin >> width;
}
void calculateArea() {
        cout << "Area: " << (length * width) << endl;
   }
};

int main() {
// 4-Create object

Rectangle rect1;

//5-Set Attribute Values(assign or read by user)

rect1.readData();
//6. Call Methods

rect1. calculateArea() ;
   }
```

Example3 calculate the area of a rectangle using a class. The program should:
read input values in main function .

```
#include <iostream>
using namespace std;
//1-Declare Class
class Rectangle {
public:

//2-Define Attributes

   double length;
   double width;
   // 3-Define Methods
   void calculateArea() {
        cout << "Area: " << (length * width) << endl;
   }};

int main() {
   //4- Create fobject
   Rectangle rect1;
// 5-Read data
   cout << "Enter length of rectangle: ";
      cin >> rect1.length;
      cout << "Enter width of rectangle: ";
      cin >> rect1.width;

   // 6-CALL methods
   rect1.calculateArea();
   return 0;}
```

Q.Reorder the comments to make the program understandable the steps of General Steps for Creating a Class and its objects

```
#include <iostream>
using namespace std;
//1
class Rectangle {
public:
//2-
   double length;
   double width;
//3-
     void readData() {
       cout << "Enter length of rectangle: ";
       cin >> length;
       cout << "Enter width of rectangle: ";
       cin >> width;
   }
void calculateArea() {
         cout << "Area: " << (length * width) << endl;
}};
int main() {
   //4
   Rectangle rect1;
  // 5
rect1.readData();
 //6
rect1.calculateArea();
   return 0;
}
```

Declare Class

Create object

Set data

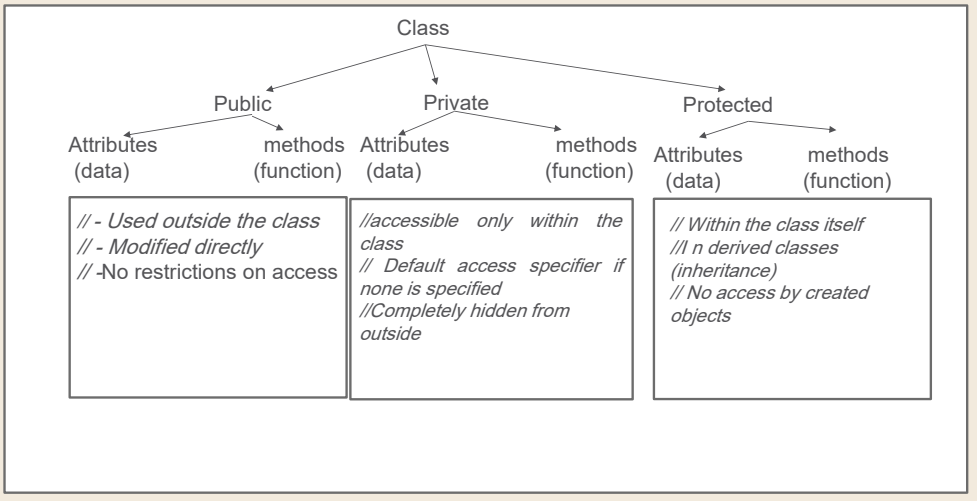
define methods

define attributes

call methods

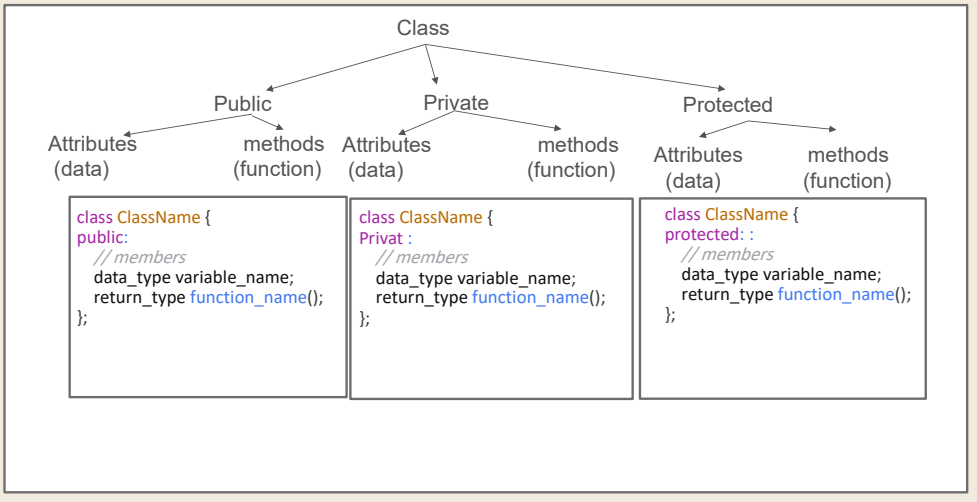
Access specifiers

The keywords **public, private, and protected** are called access specifiers. They control how class members can be accessed from different parts of the program.



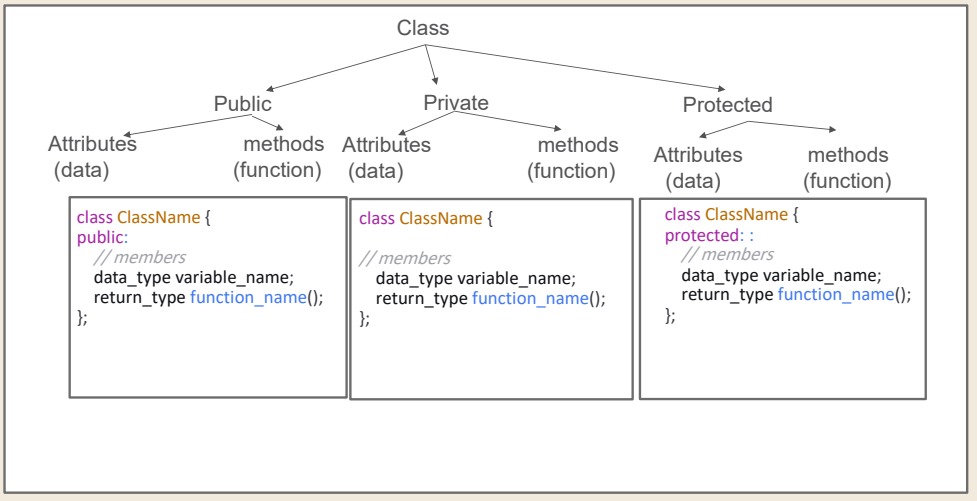
Access specifiers

The following diagram show Class Structure with Access Specifiers



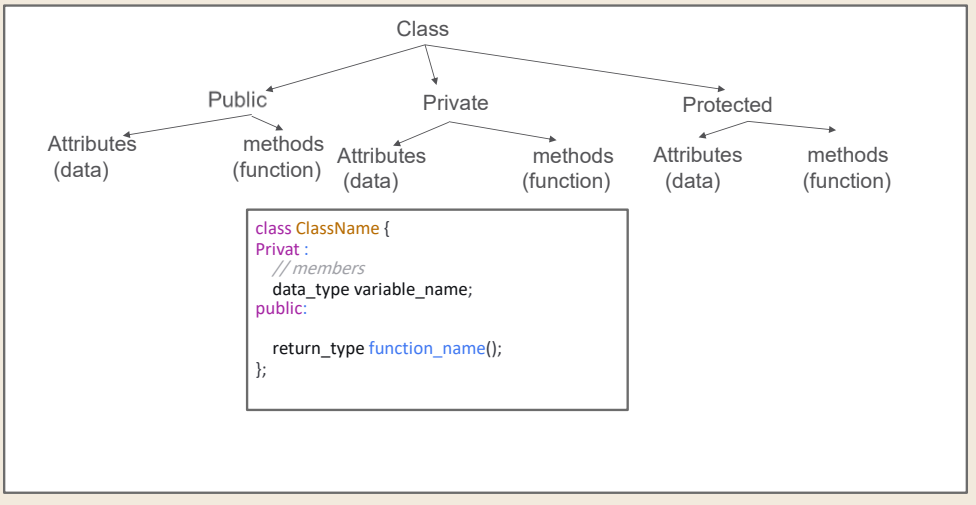
Access specifiers

The following diagram show Class Structure with Access Specifiers



Access specifiers

A class can have multiple public, protected, or private labeled sections



Question 1 : write the Key Points with ✗ or ✓

Access Type	Class Itself	Derived Classes	Outside Objects
private	✓ Yes	✗ No	✗ No
protected	✓ Yes	✓ Yes	✗ No
public	✓ Yes	✓ Yes	✓ Yes

Exampl4:Write a program to read and write two numbers using class(create one object the data(attributes are private and Methodes are public):

```
#include <iostream>
using namespace std;

class Twonumbers
{
private:
   int a, b;
public:
   void set_data(int d1, int d2)
   {
      a = d1;
      b = d2;
   }
   void show_data()
   {
      cout << "Data are " << a << "\t" << b << endl;
   }
};

int main()
{
  Twonumbers  obj;
   obj.set_data(5, 10);
   obj.show_data();

   return 0;
}
```

Private: a and **b** are private and cannot be accessed directly from outside the class.

Public : set_data() and show_data() are public and can be called from anywhere..

Exampl4:Write a program to read and write two numbers using class(create one object the data(attributes are private and Methodes are public):

```
#include <iostream>
using namespace std;

class Twonumbers
{
private:
   int a, b;
public:
   void set_data(int d1, int d2)
   {
      a = d1;
      b = d2;
   }
   void show_data()
   {
      cout << "Data are " << a << "\t" << b << endl;
   }
};

int main()
{
   Twonumbers  Obj;
Obj.a=5;
Obj.b=10;
 //  obj.set_data(5, 10);
   obj.show_data();

   return 0;
}
```

Is it correct? Why? **No.Because we use private data out side class .**

Example5:Write a program to read and write two numbers using class(create two objects the data(attributes are private and methods are public):

```
#include <iostream>
using namespace std;

class Twonumbers
{
private:
   int a, b;
public:
   void set_data(int d1, int d2)
   {
      a = d1;
      b = d2;
   }
   void show_data()
   {
      cout << "Data are " << a << "\t" << b << endl;
   }
};

int main()
{
   Twonumbers obj, obj2;
   obj.set_data(5, 10);
   obj2.set_data(2, 7);
   obj.show_data();
   obj2.show_data();

   return 0;
}
```

Private: a and b are private and cannot be accessed directly from outside the class.

Public : set_data() and show_data() are public and can be called from anywhere..

Example6:Write a C++ program that read two numbers from the user, read them in the class, and then display their sum and multiplication

```
#include <iostream>
using namespace std;

class num {
private:
   int a, b;
public:
   void read_data() {
      cout << "Enter two numbers: ";
      cin >> a >> b;
   }

   void show_results() {
      cout << "The sum of numbers = " << a + b << endl;
      cout << "The multiplication of numbers = " << a * b << endl;
   }
};

int main() {
   num obj;
   obj.read_data();
   obj.show_results();
   return 0;
}
```

Question1: Write a C++ program to calculate Heron's formula. Create a class with private attributes and public methods

$$A = \sqrt{s(s-a)(s-b)(s-c)} \qquad s = \frac{a+b+c}{2}$$

Attributes: Three variables (a,b,c), semi-perimeter(s), and area(**store data**)
Methods: Function to set data , function to calculate area and display result

.

```
#include <iostream.>
using namespace std;

#include <math.h>
class triangle
{
private :
   float a,b,c,s,area;
public:
   void set_data(float a1,float b1,float c1)
   {
   a=a1;
   b=b1;
   c=c1;
   s=(a1+b1+c1)/2;
   area=sqrt(s*(a*a)*(a-b)*(a-c));
   }
   void show_data()
   {
   cout<<"the area of triangle = "<<area<<endl;
   }
};

int main()
{
   triangle obj;
   float n1,n2,n3;
   cin>>n1>>n2>>n3;
   obj.set_data(n1,n2,n3);
   obj.show_data();
}
```